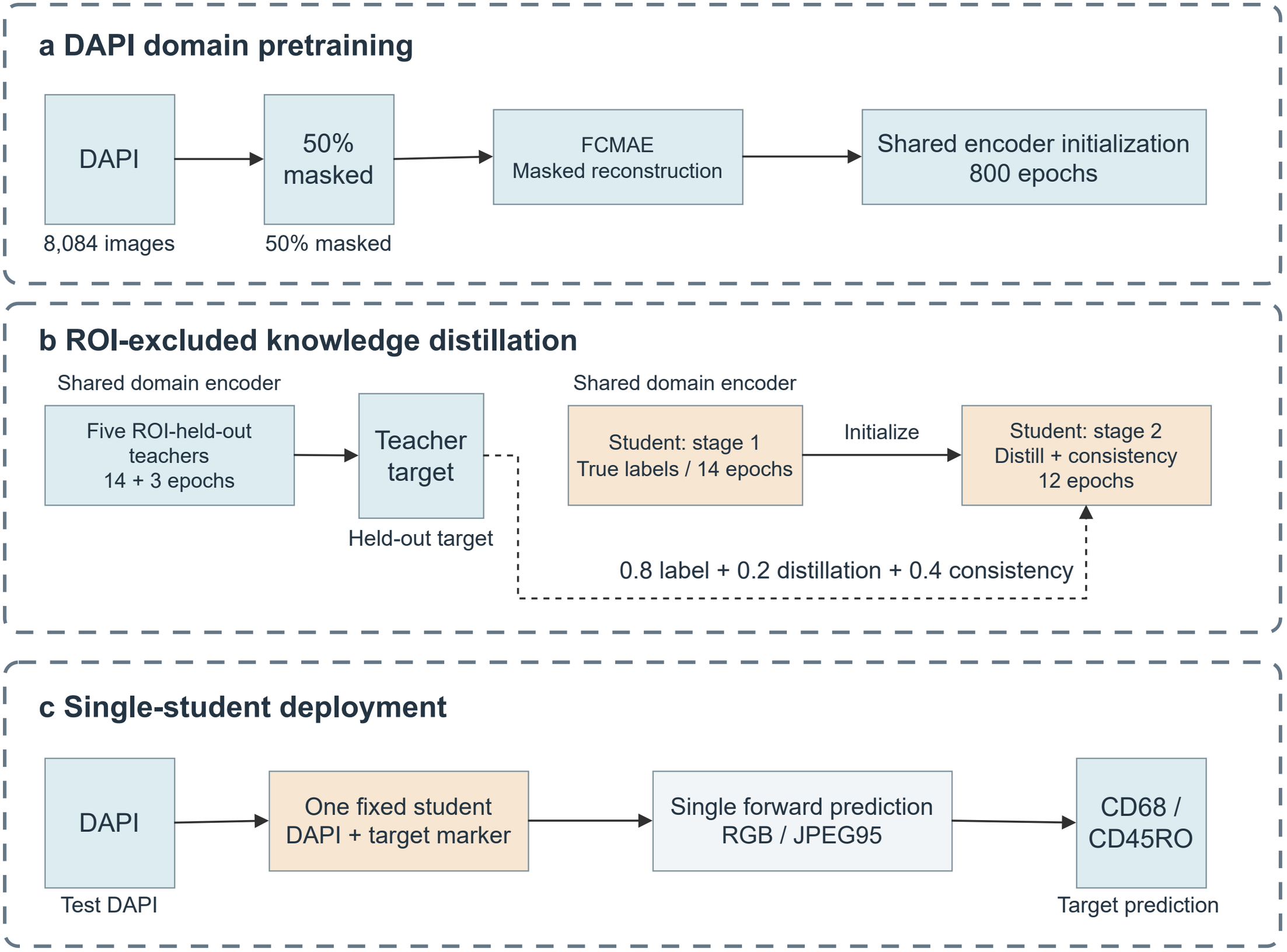
a DAPI domain pretraining
DAPI
8,084 images
50% masked
50% masked
FCMAE
Masked reconstruction
Shared encoder initialization
800 epochs
b ROI-excluded knowledge distillation
Shared domain encoder
Five ROI-held-out teachers
14 + 3 epochs
Teacher target
Held-out target
Shared domain encoder
Student: stage 1
True labels / 14 epochs
Initialize
Student: stage 2
Distill + consistency
12 epochs
0.8 label + 0.2 distillation + 0.4 consistency
c Single-student deployment
DAPI
Test DAPI
One fixed student
DAPI + target marker
Single forward prediction
RGB / JPEG95
CD68 / CD45RO
Target prediction